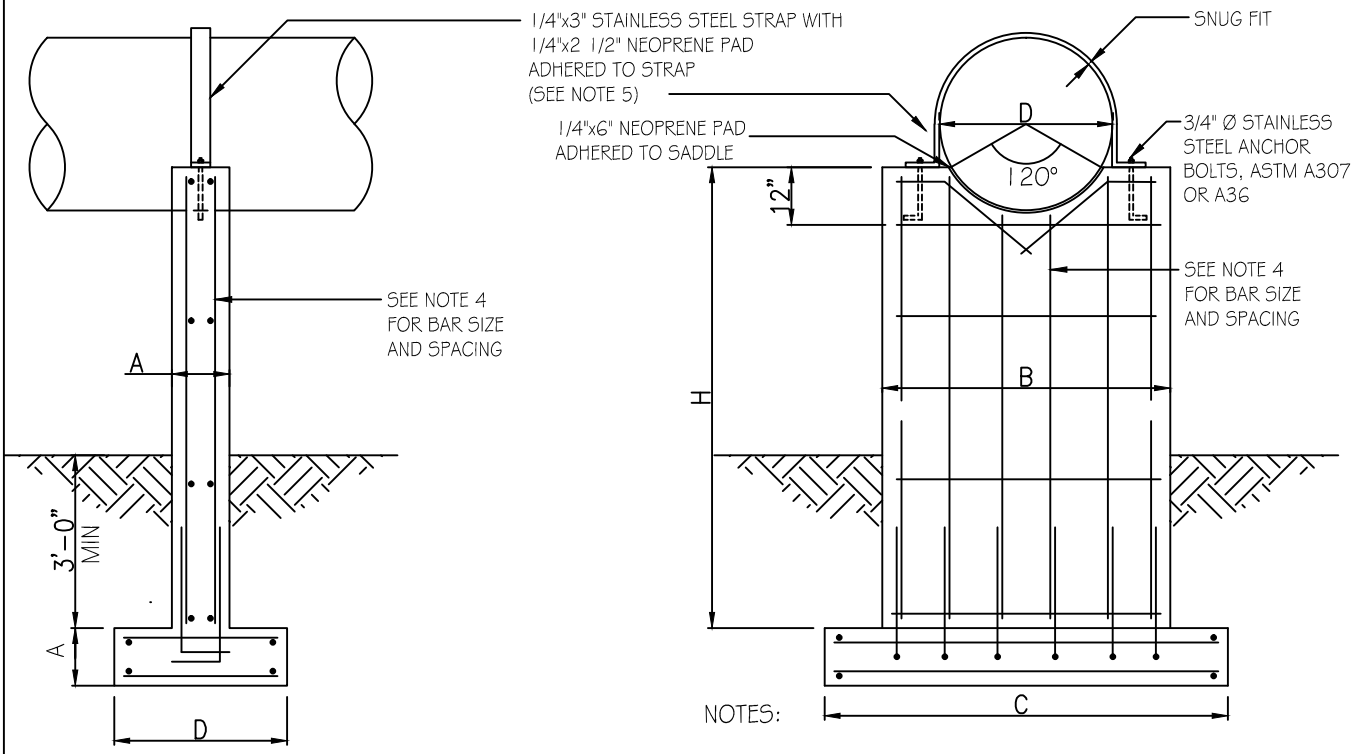


CASING PIPE DIA. "D" (IN.)	"H" (FT.)	THICKNESS "A" (IN.)	PIER WIDTH "B" (FT.)	FOOTING LENGTH "C" (FT.)	FOOTING WIDTH "D" (FT.)
6-12	≤6	12	2'-4"	5'-6"	3'-0"
	8	12	2'-4"	6'-3"	3'-0"
	10	12	2'-4"	6'-8"	3'-0"
	12	12	2'-4"	7'-2"	3'-0"
14-20	≤6	12	3'-0"	8'-0"	3'-0"
	8	12	3'-0"	9'-0"	3'-0"
	10	12	3'-0"	9'-10"	3'-0"
	12	14	3'-0"	10'-6"	3'-0"
22-28	≤6	14	3'-8"	8'-9"	4'-0"
	8	14	3'-8"	10'-0"	4'-0"
	10	14	3'-8"	11'-0"	4'-0"
	12	14	3'-8"	11'-10"	4'-0"
30-36	≤6	18	4'-4"	9'-0"	4'-0"
	8	18	4'-4"	10'-6"	4'-0"
	10	18	4'-4"	11'-6"	4'-0"
	12	18	4'-4"	12'-4"	4'-0"
38-48	≤6	18	5'-4"	9'-6"	5'-0"
	8	18	5'-4"	11'-0"	5'-0"
	10	18	5'-4"	12'-0"	5'-0"
	12	18	5'-4"	12'-10"	5'-0"
51-56	≤6	18	6'-4"	9'-10"	5'-0"
	8	18	6'-4"	11'-4"	5'-0"
	10	18	6'-4"	12'-4"	5'-0"
	12	18	6'-4"	13'-2"	5'-0"

NOTES:

1. SHALLOW FOUNDATION DESIGN SHOWN ON THIS DETAIL IS BASED ON THE FOLLOWING PARAMETERS:
   - ALLOWABLE SOIL BEARING CAPACITY = 2000 PSF
   - CONCRETE COMPRESSIVE STRENGTH = 4000 PSI
   - GRADE 60 REINFORCING STEEL
   - MAXIMUM STREAM VELOCITY = 10 FT/SEC
   - MAXIMUM SUPPORT HEIGHT (H) = 12'-0"

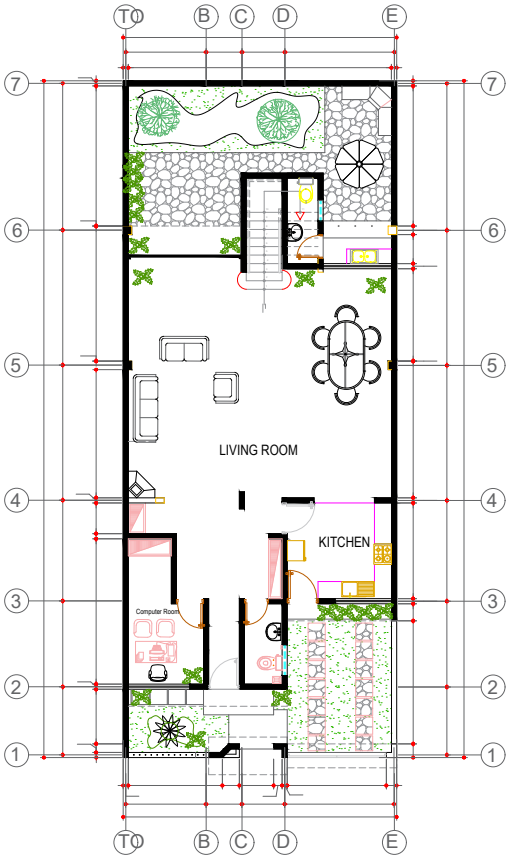
   IF FIELD CONDITIONS REQUIRE ANY DEVIATION FROM THESE PARAMETERS, THE FOUNDATION DESIGN SHALL BE REVIEWED BY THE ENGINEER.
2. IF SUBGRADE AT LOCATION OF SUPPORTS IS DEEMED UNABLE TO WITHSTAND 2000 PSF BEARING PRESSURE, A PILE SUPPORTED FOUNDATION SHALL BE UTILIZED AS PER DRAWING S-15.
3. IF BEDROCK IS ENCOUNTERED WHICH WILL PREVENT 3-FEET MINIMUM COVER OVER FOOTING, DOWELS SHALL BE DRILLED INTO BEDROCK PRIOR TO PLACING FOUNDATION. SEE DRAWING S-17.
4. TWELVE-INCH AND FOURTEEN-INCH THICK PIERS AND FOOTINGS SHALL BE REINFORCED WITH #5 BARS AT 12 INCHES OC IN EACH DIRECTION ON EACH FACE. EIGHTEEN-INCH WIDE PIERS AND FOOTINGS SHALL BE REINFORCED WITH #7 BARS AT 12 INCHES OC IN EACH DIRECTION ON EACH FACE.
5. EIGHTEEN-INCH THICK PIERS SHALL REQUIRE TWO STRAPS OVER THE PIPE INSTEAD OF ONE (AS SHOWN).
6. WHEN CONCRETE SUPPORTS ARE REQUIRED TO BE LOCATED WITHIN A STREAM AND ARE NOT COVERED WITH BACKFILL, SEE DRAWING S-19 FOR MODIFICATIONS TO UPSTREAM FACE OF SUPPORT.

CITY OF RALEIGH

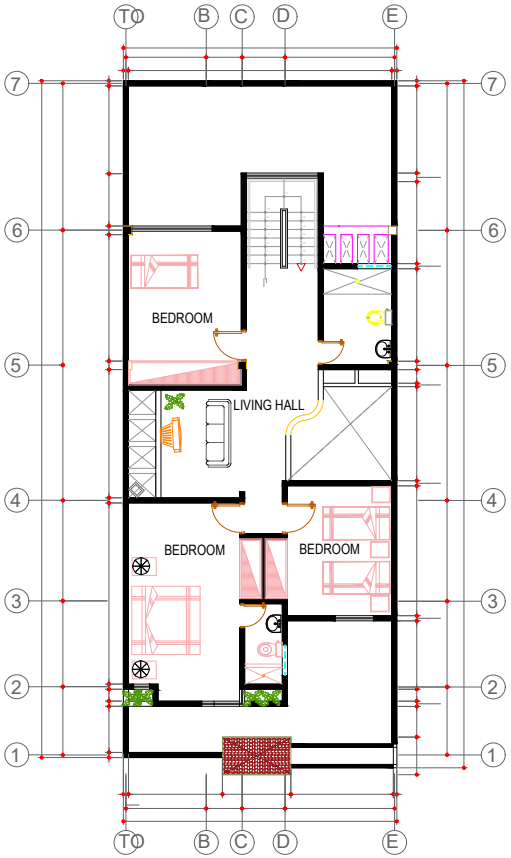
DEPARTMENT OF PUBLIC UTILITIES

AERIAL PIPE CROSSING CONCRETE PIER DETAIL

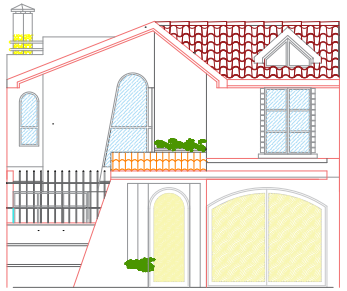
DWG. NO.	REVISIONS	DATE	REVISIONS	DATE
S-14	D.H.L.	6/16/08		



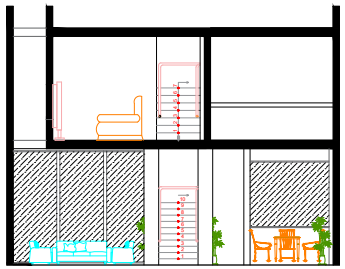
GROUND FLOOR PLAN



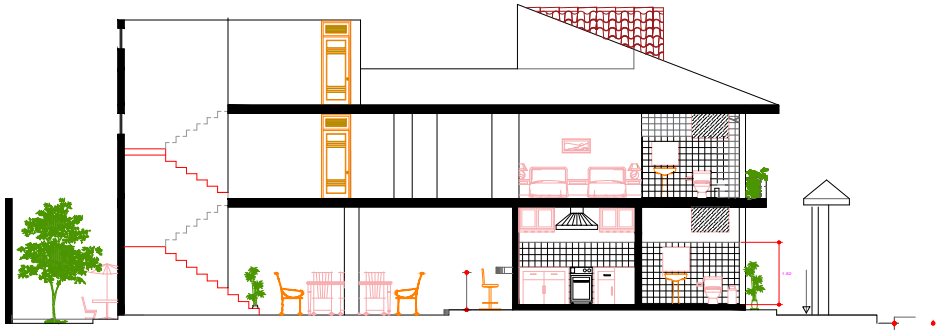
FIRST FLOOR PLAN



FRONT ELEVATION

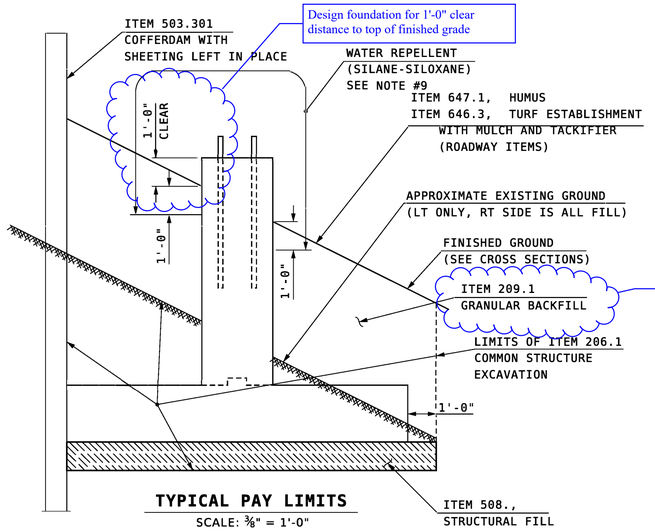


SECTION DETAIL



SECTION DETAIL

SIGN FOOTING MASONRY & REINFORCEMENT FOR ITEM NO. 615.10001



TYPICAL PAY LIMITS

SCALE: ⅜" = 1'-0"

SIGN FOOTING NOTES

1. FOR THE INSTALLATION AND PRETENSIONING OF ANCHOR RODS, SEE SPECIAL PROVISION AMENDMENT TO SECTION 615, TRAFFIC SIGNS.
2. THERE ARE 2 FOOTINGS OF THIS CONFIGURATION REQUIRED AND THE QUANTITIES GIVEN ARE TOTALED FOR BOTH FOOTINGS.
3. ALL REINFORCING STEEL SHALL HAVE 3" MINIMUM CLEAR COVER FROM CONCRETE SURFACES.
4. FOOTINGS SHALL BE BACKFILLED TO FINAL GROUND BEFORE THE SIGN SUPPORT STRUCTURE IS INSTALLED.
5. FINAL GROUND ELEVATIONS SHALL BE TAKEN FROM THE ROADWAY CROSS SECTIONS. IF ACTUAL FINAL GROUND ELEVATIONS DIFFER BY MORE THAN 1'-0", THEN A COMPLETE REDESIGN OF THE SIGN FOOTING WILL BE REQUIRED.
6. THE LEFT FOOTING DESIGN SIGN AREA IS 442.4 SF. THE RIGHT FOOTING DESIGN SIGN AREA IS 504.0 SF. BOTH DESIGN SIGN AREAS INCLUDED A 30% INCREASE.

Edit note to designed area.

7. STAINLESS STEEL STD. GR. WIRE CLOTH, ¼" MAX. OPENING WITH MIN. WIRE DIA. OF AWG NO. 16 WITH 2" LAP. SECURE WITH ¾" STAINLESS STEEL BANDING AFTER ANCHOR RODS ARE FULLY TIGHTENED AND TESTED. (SUBSIDIARY TO ITEM 615.XX)
8. FOOTING SUPPORTED ON STRUCTURAL FILL ON UNDISTURBED SOIL WITH A NOMINAL BEARING CAPACITY OF 15.1 KSF IN COMBINATION WITH A RESISTANCE FACTOR OF 0.45 (STRENGTH). SEE THE GEOTECHNICAL REPORT FOR FURTHER INFORMATION.

Include note from Geotechnical Report noting soil capacity and any requirements.

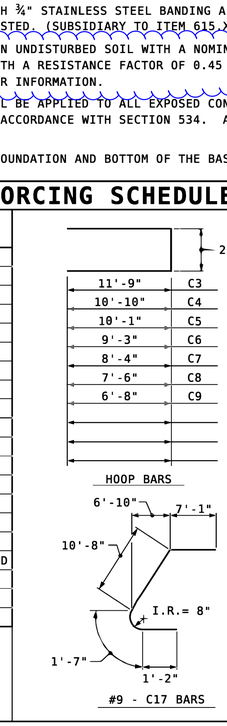
9. WATER REPELLENT (SILANE-SILOXANE) SHALL BE APPLIED TO ALL EXPOSED CONCRETE SURFACES TO 1'-0" BELOW FILL LINES IN ACCORDANCE WITH SECTION 534. ALL COSTS SUBSIDIARY TO ITEM 520.2.
10. NO GROUT SHALL BE PLACED BETWEEN THE FOUNDATION AND BOTTOM OF THE BASE PLATE.

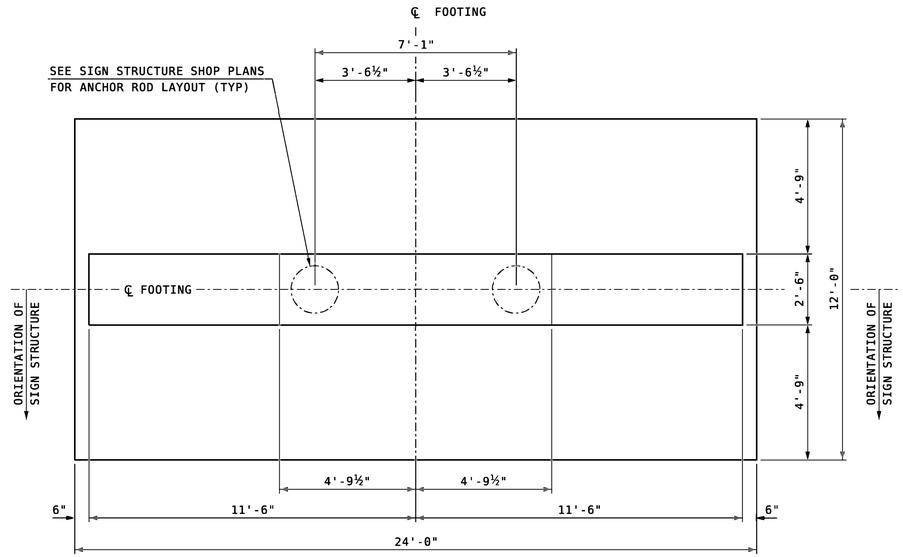
SIGN FOOTING REINFORCING SCHEDULE

MARK	SIZE	NO.	UNBENT LENGTH	TYPE
C1	#7	48	23'-6"	STRAIGHT
C2	#7	96	11'-6"	STRAIGHT
C3	#4	4	25'-6"	HOOP
C4	#4	4	23'-8"	HOOP
C5	#4	4	22'-2"	HOOP
C6	#4	4	20'-6"	HOOP
C7	#4	4	18'-8"	HOOP
C8	#4	4	17'-0"	HOOP
C9	#4	4	15'-4"	HOOP
C10	#7	40	9'-6"	STRAIGHT
C11	#4	8	2'-6"	STRAIGHT
C12	#4	8	3'-8"	STRAIGHT
C13	#4	8	4'-10"	STRAIGHT
C14	#4	8	6'-1"	STRAIGHT
C15	#4	8	7'-3"	STRAIGHT
C16	#4	8	8'-6"	STRAIGHT
C17	#9	16	20'-6"	BENT & CURVED

REINFORCING STEEL IN SCHEDULE IS FOR 2 SIGN FOOTINGS.
TOTAL WEIGHT OF REINFORCING STEEL IN TABLE IS 7012 lb.

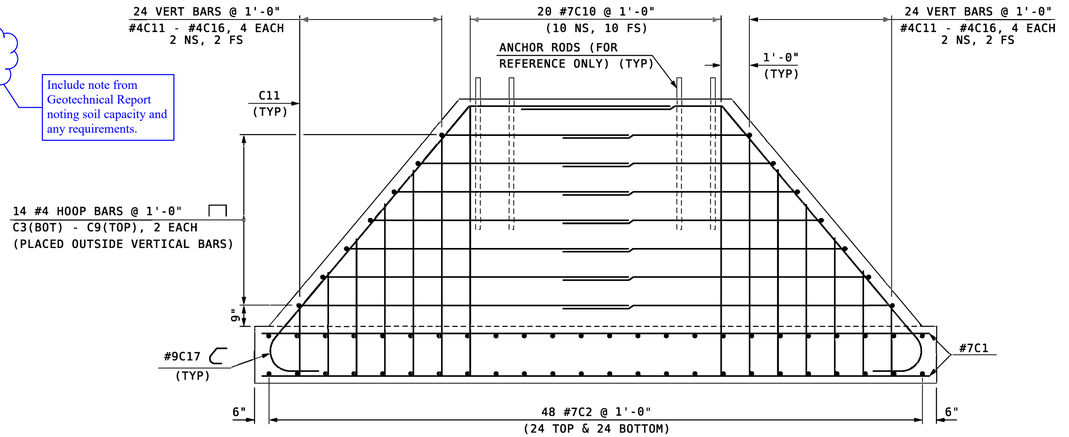
REINFORCING STEEL SHALL BE GRADE 60





SIGN FOOTING MASONARY PLAN

SCALE: ⅜" = 1'-0"



REINFORCING - SIDE ELEVATION

SCALE: ⅜" = 1'-0"

Quantify Item 209.1 for all projects

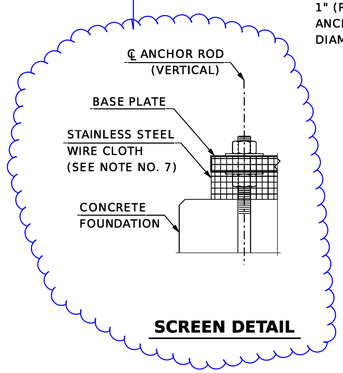
QUANTITY TABLE (2 FOOTINGS)

ITEM NO.	ITEM DESCRIPTION	QUANTITY	UNIT
206.1	COMMON STRUCTURE EXCAVATION	85	CY
209.1	GRANULAR BACKFILL	170	CY
503.301	COFFERDAMS WITH SHEETING	2	U
	LEFT IN PLACE		
508	STRUCTURAL FILL	37	CY
520.2	CONCRETE CLASS B	66.8	CY
544.1	REINFORCING STEEL (ROADWAY)	7012	LB

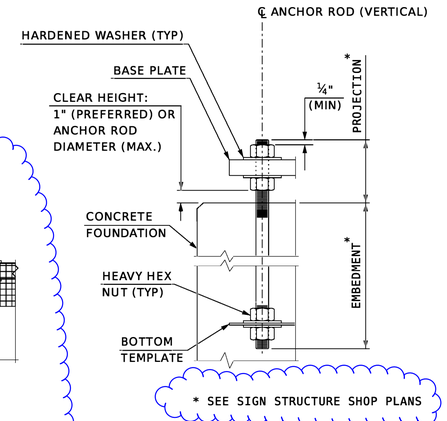
SIGN FOOTING ELEVATION TABLE

STATION	OFFSET	ELEV "A"	ELEV "B"
172+00	LT 80	261	251
172+00	RT 80	258	248

To be included and installed on all foundations (even without wires) to help prevent debris covering anchors

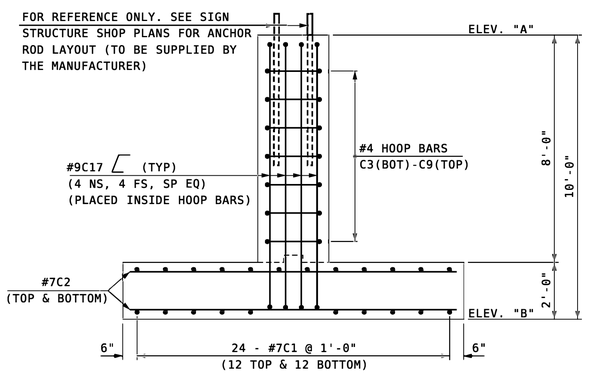


SCREEN DETAIL



ANCHOR ROD DETAIL

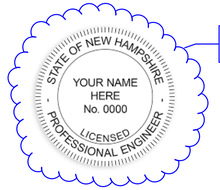
Add note



REINFORCING - END ELEVATION

SCALE: ⅜" = 1'-0"

Place a NH PE stamp and sign



Note the inventory number of the structure/foundation obtained from Bureau of Traffic

INV #: 389-SPTPK-0015-NS20.9

Note the specific location of foundation obtained from Bureau of Traffic

SAMPLE PLAN

DATE: 12/2024

STATE OF NEW HAMPSHIRE
DEPARTMENT OF TRANSPORTATION * BUREAU OF BRIDGE DESIGN

TOWN ROCHESTER BRIDGE NO. N/A STATE PROJECT 10620L

LOCATION NH ROUTE 16 - 400' SOUTH OF EXIT 15 NB OFF RAMP

OVERHEAD SIGN STRUCTURE FOOTING

REVISIONS AFTER PROPOSAL	BY	DATE

	BY	DATE		BY	DATE
DESIGNED			CHECKED		
DRAWN			CHECKED		
QUANTITIES			CHECKED		
ISSUE DATE			FEDERAL PROJECT NO.		
REV. DATE			----------		

SUBDIRECTORY	.DGN LOCATOR	SHEET SCALE
ROCHESTER/10620L	615.1000X	AS NOTED

BRIDGE SHEET OF
FILE NUMBER
SHEET NO.
TOTAL SHEETS